

Wenhao Peng

RF Acoustic Wave Devices

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Biography

Wenhao Peng is a Staff Electrical Engineer specializing in RF acoustic-wave technology at Skyworks Solutions, working across device modeling, filter design, and high-volume manufacturing of bulk acoustic-wave resonators and filters. His work connects first-principles models of material and acoustic behavior with practical resonator and filter design. In ferroelectric BAW technology, he developed a physics-based equivalent-circuit framework that bridges nonlinear ferroelectric thermodynamics and measured resonator behavior. In piezoelectric FBAR technology, he established an analytical relation between field-based and resonance-frequency-based definitions of electromechanical coupling, making modal field distributions directly useful for evaluating layer stacks and device performance. He is now extending modal methods to reduce the computational cost of finite element acoustic simulation for production filter design. He also demonstrated the first FBAR operating above 50 GHz using a periodically poled AlN/ScAlN/AlN trilayer. He received the Ph.D. degree in electrical and computer engineering from the University of Michigan, Ann Arbor, in 2026.

Selected Contributions

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| Modal analysis | Proved the analytical relation between the two standard definitions of electromechanical coupling (k_t^2): the Berlincourt form computed from acoustic field distributions, and the form determined from the series and parallel resonance frequencies. Derived the relation through modal analysis of the open circuit eigenmodes of an FBAR, filling a gap in the literature. The result makes acoustic field plots quantitatively interpretable, showing what a given mode contributes to resonator coupling, and enables direct evaluation of modal contributions to layer stack performance, reducing reliance on iterative optimization against simulated admittance. This relation was applied in both mm-Wave FBAR papers, with the derivation presented at IMS 2024 and IMS 2025, and is now being extended to reduce finite element simulation cost in production filter design (see Experience). |
| Finite element methods | Adapted interpolation methods from finite element analysis into the thermodynamic modeling of ferroelectrics, bridging a bulk thermodynamic description and a compact circuit representation. Produced the first equivalent circuit model of ferroelectric driven BAW resonators derived from first principles rather than empirical fitting, reproducing measured response across DC bias. First author, IEEE TMTT 2022 and IMS 2021. |
| mm-Wave scaling | First FBAR demonstrated above 50 GHz: a 56 GHz periodically poled AlN/ScAlN/AlN trilayer, extending demonstrated FBAR operation to more than five times the frequency of the prior state of the art, which was below 10 GHz. A second generation 36 GHz device applied the modal analysis result above to electrode and layer stack selection. First author on both (IMS 2024, IMS 2025). Built on co-demonstrated controlled ferroelectric polarization switching in ultrawide bandgap AlN/ScAlN multilayers (Applied Physics Letters 2023). |

Switchless frontends Co-developed the first switchless quad band filter bank based on ferroelectric BST FBARs. Related work produced the first intrinsically switchable balanced ferroelectric FBAR filter and the first RF acoustic phase inversion switch based on poled ferroelectrics.

Experience

- Feb 2026–Present **RF Acoustic Staff Electrical Engineer**, *Skyworks Solutions, Inc.*, Irvine, CA.
- Serve as a technical bridge across device modeling, filter design, and high volume manufacturing, applying equivalent circuit and multiphysics modeling to guide design decisions and troubleshoot performance throughout the design to production pipeline.
 - Contribute to resonator stack design and optimization for higher frequency cellular bands, balancing performance, tunability, and manufacturability requirements.
 - Extend the modal analysis methods developed during doctoral research to reduce the computational cost of finite element acoustic simulation, with the goal of making mask level response prediction deployable inside the filter design flow, where full finite element simulation had previously been limited by runtime and computational cost (ongoing R&D).
 - Work with modeling, process, and production teams to align design targets with fab capability and yield.
- Sep 2019–Dec 2025 **Graduate Student Research Assistant**, *University of Michigan*, Ann Arbor, MI.
Advisor: Prof. Amir Mortazawi
- 2025 *mm-Wave acoustic resonators based on periodically poled III-nitride thin films*
- Demonstrated the first mm-Wave FBARs using AlN/ScAlN/AlN periodically poled trilayers, including the first such device operating above 50 GHz, at 56 GHz, followed by a second generation 36 GHz resonator with a redesigned electrode and layer stack.
 - Proved the Berlincourt to resonance frequency coupling relation described above as the theoretical groundwork for the device work, providing direct evaluation of modal contributions and reducing reliance on iterative optimization against admittance responses.
 - Advanced the nanofabrication and pulsed field polarization switching processes, including current response analysis.
 - Performed mm-Wave characterization to 67 GHz with on-wafer de-embedding structures validated in ADS Momentum and Ansys HFSS.
- Reconfigurable RF frontends based on intrinsically switchable BST acoustic resonators*
- Developed the first equivalent circuit model of ferroelectric driven BAW resonators grounded in ferroelectric thermodynamics. Adapted finite element interpolation methods to bridge the bulk thermodynamic description and a compact circuit representation, capturing the underlying physics rather than empirically fitting measured responses. Validated the model by direct comparison with measurements across DC bias.
 - Contributed to the first switchless quad band BST FBAR filter bank and the first intrinsically switchable balanced ferroelectric FBAR filter at 2 GHz.
 - Contributed to the first RF acoustic phase inversion switch based on poled ferroelectric stacked crystal filters.
 - Synthesized filters using the image parameter method, applying Mason and mBVD resonator models, and carried the designs through complete cleanroom fabrication and DC bias dependent characterization.

Education

- 2026 **Ph.D., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Dissertation: *Layered Ferroelectric Millimeter Wave BAW Resonators: Modeling and Validation based on On-Wafer Characterization*. Advisor: Prof. Amir Mortazawi.
- 2024 **M.S.E., Mechanical Engineering**, *University of Michigan*, Ann Arbor, MI.
Dynamics and vibrations

- 2019 **M.S., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Integrated circuits and VLSI
- 2018 **B.S.E., Electrical Engineering**, *University of Michigan*, Ann Arbor, MI.
Summa Cum Laude
- 2018 **B.S.E., Electrical and Computer Engineering**, *Shanghai Jiao Tong University*, Shanghai, China.
Dual degree with U-M

Technical Skills

- Devices & materials BAW/FBAR resonator and filter design; thin film piezoelectrics and ferroelectrics (AlN, ScAlN, BST); periodically poled III-nitride multilayers; MEMS; RF frontend architectures
- Modeling Mason and mBVD equivalent circuits; thermodynamic modeling of ferroelectric nonlinearity; modal analysis; image parameter filter synthesis; multiphysics FEM (COMSOL); EM co-simulation (ADS Momentum, Ansys HFSS); structural vibration analysis
- Fabrication Cleanroom process integration; photolithography; physical vapor deposition; wet/dry/plasma etching; thermal oxidation; device processing compatible with MBE grown films; pulsed field poling
- Measurement On-wafer S-parameter characterization to 67 GHz; GSG probing; VNA calibration and de-embedding; DC bias dependent RF characterization; material parameter extraction
- Tools ADS, COMSOL Multiphysics, Ansys HFSS, MATLAB, Python, Cadence, LLM-assisted Python development, code review, numerical workflow automation, and technical documentation.

Journal Publications

- 2025 H. Desai, **W. Peng**, and A. Mortazawi, "Single-Pole Single-Throw RF Acoustic Phase Inversion Switch Leveraging Poled Ferroelectrics," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 1, pp. 6–13, 2025.
- 2023 D. Wang, P. Wang, S. Mondal, J. Liu, M. Hu, M. He, S. Nam, **W. Peng**, S. Yang, D. Wang, Y. Xiao, Y. Wu, A. Mortazawi, and Z. Mi, "Controlled Ferroelectric Switching in Ultrawide Bandgap AlN/ScAlN Multilayers," *Appl. Phys. Lett.*, vol. 123, no. 10, p. 103506, 2023.
- 2023 S. Nam, **W. Peng**, P. Wang, D. Wang, Z. Mi, and A. Mortazawi, "A mm-Wave Trilayer AlN/ScAlN/AlN Higher Order Mode FBAR," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 6, pp. 803–806, 2023.
- 2022 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Phenomenological Circuit Modeling of Ferroelectric-Driven Bulk Acoustic Wave Resonators," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 1, pp. 919–925, 2022.
- 2021 S. Nam, M. Z. Koohi, **W. Peng**, and A. Mortazawi, "A Switchless Quad Band Filter Bank Based on Ferroelectric BST FBARs," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 6, pp. 662–665, 2021.

Conference Publications

- 2025 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 36 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, San Francisco, CA, 2025, pp. 786–789. *(presented by author)*
- 2024 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 56 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, Washington, DC, 2024, pp. 150–153. *(presented by author)*
- 2021 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, “Physics Based Modeling of Electrostriction Based BAW Resonators,” *IEEE MTT-S International Microwave Symposium (IMS)*, Atlanta, GA, 2021, pp. 214–217. *(presented by author)*
- 2020 M. Z. Koohi, **W. Peng**, and A. Mortazawi, “An Intrinsically Switchable Balanced Ferroelectric FBAR Filter at 2 GHz,” *IEEE/MTT-S International Microwave Symposium (IMS)*, 2020, pp. 131–134.
- 2018 Y. Dai, **W. Peng**, Y. Wang, L.-X. Chuo, K. Suri, H. Zheng, D. Wentzloff, and H.-S. Kim, “Implementation and Evaluation of Bi-Directional WiFi Back-Channel Communication,” *IEEE 29th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2018, pp. 1–7.

Peer Review Contributions

- 47 reviews **IEEE:** J. Microelectromechanical Systems (8), Microwave and Wireless Technology Letters (6), Electron Device Letters (4), J. Microwaves (1). **APS:** Physical Review Letters (5), Physical Review Applied (2), Physical Review Materials (2), Physical Review A (1), Physical Review B (1). **Other:** Physics of Fluids (1), Micromachines (10), Photonics (2), Sensors (1), Microwave (MDPI) (2), Progress in Quantum Electronics (1).

Selected Graduate Coursework

Wave Propagation in Elastic Solids; Advanced Vibrations of Structures; Advanced Engineering Acoustics; Theory of Solid Continua; Math Methods in ME; Advanced Solid State Microwave Circuits; Electromagnetics Theory; Introduction to MEMS; Monolithic Amplifier Circuits; Analog Integrated Circuits; VLSI Design; Computer Architecture; Digital Communication Signals and Systems; Probabilistic Methods in Engineering